Linear Transformations and Multivariate RVs

L20

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CSE400: Fundamentals of Probability in Computing

Lecture 20: Linear Transformations and Expectations of Multiple RVs Instructor: Dhaval Patel, PhD

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1. Statistical Representation of Random Vectors

For multiple random variables $X_1, X_2, \ldots, X_N$, we represent them compactly as a **Random Vector** $\underline{X}$:

$$\underline{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix} \quad [\text{cite: } 23]$$

Mean Vector

The mean vector $\underline{\mu}_X$ represents the average value of the random vector. Each element in the vector corresponds to the mean of an individual random variable:

$$\underline{\mu}_X = E[\underline{X}] = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix} \quad \text{[cite: 32, 42, 46]}$$

Correlation Matrix (R_{XX})

The correlation matrix represents the correlation between multiple random variables. It is defined as:

$$R_{XX} = E[\underline{X}\,\underline{X}^T]$$

- **Element Representation:** $R_{XX}(i,j) = E[X_i X_j]$ for $i, j \leq N$.
- **Structure:** It is an $N \times N$ symmetric matrix.

Covariance Matrix (C_{XX})

The covariance matrix acts as a "relationship map" for random vectors. It is defined as:

$$C_{XX} = E[(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T]$$

- **Practical Structure:** The diagonal elements are variances, and off-diagonal elements are covariances:

$$C_{XX} = \begin{bmatrix} Var(X_1) & Cov(X_1, X_2) & \dots \\ Cov(X_2, X_1) & Var(X_2) & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

- **Property:** It is symmetric because $Cov(X_i, X_j) = Cov(X_j, X_i)$.

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2. Linear Transformations of Random Vectors

Consider the transformation $\underline{Y} = \underline{AX} + \underline{b}$, where:

- $\underline{A}$: Transformation matrix (scales and rotates data).
- $\underline{b}$: Constant vector (shifts/translates the center).
- $\underline{X}$: Input random vector.

Mapping Properties

Property	Transformed Result
Mean	$\mu_V = A\mu_X + b$
Covariance	$C_{VV} = AC_{XX}A^T$

Property	Transformed Result
Correlation	$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$

Derivations

- **Mean:** $\mu_Y = E[\underline{AX} + \underline{b}]$. By linearity, this becomes $E[\underline{AX}] + E[\underline{b}]$, resulting in $\underline{A}\mu_X + \underline{b}$.
- **Covariance:** The deviation from the mean $(\underline{Y} - \mu_Y)$ is independent of the shift $\underline{b}$, as $(\underline{AX} + \underline{b}) - (\underline{A}\mu_X + \underline{b}) = \underline{A}(\underline{X} - \mu_X)$. Thus, $C_{YY} = E[\underline{A}(\underline{X} - \mu_X)(\underline{X} - \mu_X)^T \underline{A}^T] = \underline{A}C_{XX}\underline{A}^T$.

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3. Multivariate Gaussian Distributions

1D and 2D Cases

- **1D (Univariate):** $f_X(x) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.
- **2D (Joint):** Defined for two variables X and Y using means, variances, and correlation coefficient ρ_{XY}.

N-Dimensional Joint PDF

For an N-dimensional random vector $\underline{X}$ with mean $\underline{\mu}_X$ and covariance matrix C_{XX}:

$$f_{\underline{X}}(x) = \frac{1}{(2\pi)^N \det(C_{XX})} \exp\left(-\frac{1}{2}(x - \underline{\mu}_X)^T C_{XX}^{-1}(x - \underline{\mu}_X)\right) \text{ [cite: 324]}$$

Special Case: Uncorrelated Gaussian Variables

If N jointly Gaussian variables are mutually uncorrelated, then $Cov(X_i, X_j) = 0$ for $i \neq j$.

- The covariance matrix C_{XX} is diagonal.
- The joint PDF factorizes into the product of individual Gaussian PDFs:

$$f_{\underline{x}}(\underline{x}) = \prod_{n=1}^{N} \frac{1}{2\pi\sigma_n^2} \exp\left(-\frac{(x_n - \mu_n)^2}{2\sigma_n^2}\right)$$

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4. Worked Examples

Example: Numerical Linear Transformation

Given $\underline{Y} = \underline{AX}$ with:

$$\underline{\mu}_X = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \underline{A} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}, \underline{R}_{XX} = \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix}.$$

- **(a) Mean vector of Y:** $\underline{\mu}_Y = \underline{A}\underline{\mu}_X = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}.$
- **(b) Correlation matrix of Y:** $\underline{R}_{YY} = \underline{A}\underline{R}_{XX}\underline{A}^T = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}.$
- **(c) Covariance matrix of Y:** $\underline{C}_{YY} = \underline{R}_{YY} - \underline{\mu}_Y\underline{\mu}_Y^T.$
  - Outer product $\underline{\mu}_Y\underline{\mu}_Y^T = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}.$
  - $\underline{C}_{YY} = \begin{bmatrix} 17 & -6 & -9 \\ -6 & 14 & -8 \\ -9 & -8 & 19 \end{bmatrix} - \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}.$

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5. Case Study: Smart City Traffic (Ahmedabad)

Random Vector $\underline{X}$ (N=5):

1. X_1: Traffic flow
2. X_2: Average speed
3. X_3: Rainfall intensity
4. X_4: Temperature
5. X_5: Air pollution (AQI)

Intuition from Covariance:

- $Cov(X_1, X_3) > 0$: More rain $\rightarrow$ more congestion.
- $Cov(X_2, X_3) < 0$: More rain $\rightarrow$ lower speed.
- $Cov(X_1, X_5) > 0$: More traffic $\rightarrow$ higher pollution.

Congestion Index (Y): A linear transformation $Y = 0.6X_1 - 0.4X_2 + 0.8X_3$.
If $\underline{X}$ is Gaussian, then Y is also Gaussian.

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6. Homework Assignments

1. **Finding Mean Vector:** Given C_{XX} and R_{XX}, find $\underline{\mu}_X$ using $\underline{\mu}_X\underline{\mu}_X^T = R_{XX} - C_{XX}$.
2. **Variance Proof:** Prove $Var[X+Y] = Var[X] + Var[Y] + 2Cov(X,Y)$.

Would you like me to walk through the algebraic simplification of the 2D Joint Gaussian PDF

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CSE400: Fundamentals of Probability in Computing

Lecture 20: Linear Transformations and Expectations of Multiple RVs

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1. Statistical Representation of Random Vectors

Random Vector Definition

For a set of random variables $X_1, X_2, \ldots, X_N$, we represent them as a single vector $\underline{X}$.

- **Vector Form:** $\underline{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$.
- This provides a compact representation for multiple random variables.

Mean Vector ($\underline{\mu}_X$)

The mean vector represents the average value of the random vector.

- **Definition:** $\underline{\mu}_X = E[\underline{X}]$.
- **Structure:** Each element in the vector corresponds to the mean of one individual random variable:

$$\underline{\mu}_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

Correlation Matrix (R_{XX})

Represents the correlation between multiple random variables.

- **Definition:** $R_{XX} = E[\underline{X}\underline{X}^T]$.
- **Element Representation:** $R_{XX}(i,j) = E[X_i X_j]$ for $i, j \leq N$.
- **Property:** R_{XX} is a symmetric matrix of size $N \times N$.

Covariance Matrix (C_{XX})

Often called the "relationship map" of random vectors.

- **Definition:** $C_{XX} = E[(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T]$.
- **Practical Structure:** Diagonal elements are variances, and off-diagonal elements are covariances.

$$C_{XX} = \begin{bmatrix} Var(X_1) & Cov(X_1, X_2) & \dots \\ Cov(X_2, X_1) & Var(X_2) & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

- **Property (Symmetry):** Because $Cov(X_i, X_j) = Cov(X_j, X_i)$, the matrix is a "mirror" of itself.

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2. Linear Transformations of Random Vectors

The Linear Model

Consider the transformation: $\underline{Y} = \underline{AX} + \underline{b}$.

- $\underline{A}$: Transformation matrix (scales and rotates data).
- $\underline{b}$: Constant vector (shifts/translates the center).
- $\underline{X}$: Input random vector.

Mapping Properties

Property	Transformed Result
Mean	$\mu_Y = A\mu_X + b$
Covariance	$C_{YY} = AC_{XX}A^T$

Property	Transformed Result
Correlation	$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$

Key Derivations

1. Mean Transformation (μ_Y)

1. Substitute $\underline{Y} = \underline{AX} + \underline{b}$ into the expectation: $E[\underline{AX} + \underline{b}]$.
2. Apply linearity of expectation: $E[\underline{AX}] + E[\underline{b}]$.
3. Result: $\underline{A}\mu_X + \underline{b}$.

2. Covariance Transformation (C_{YY})

1. **Observation:** Deviation from the mean is independent of shift $\underline{b}$:

   $\underline{Y} - \mu_Y = (\underline{AX} + \underline{b}) - (\underline{A}\mu_X + \underline{b}) = \underline{A}(\underline{X} - \mu_X)$.
2. Apply definition: $C_{YY} = E[(\underline{Y} - \mu_Y)(\underline{Y} - \mu_Y)^T]$.
3. Substitute deviation: $C_{YY} = E[\underline{A}(\underline{X} - \mu_X)(\underline{X} - \mu_X)^T \underline{A}^T]$.
4. Result: $C_{YY} = \underline{A}C_{XX}\underline{A}^T$. **Note:** Shift $\underline{b}$ does not affect spread.

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3. Multivariate Gaussian Distributions

From 1D to N-Dimensional

- **Univariate (1D):** $f_X(x) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.
- **Joint (2D):**

$$f_{XY}(x,y) = \frac{\exp\left\{-\frac{1}{2(1-\rho_{XY}^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho_{XY}\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right]\right\}}{2\pi\sigma_X\sigma_Y \quad 1-\rho_{XY}^2}$$

- **General Vector PDF (N-Dimensional):**

$$f_{\underline{X}}(x) = \frac{1}{(2\pi)^N \det(C_{XX})} \exp\left(-\frac{1}{2}(x - \underline{\mu}_X)^T C_{XX}^{-1}(x - \underline{\mu}_X)\right)$$

Special Case: Uncorrelated Variables

If N jointly Gaussian variables are mutually uncorrelated:

- $Cov(X_i, X_j) = 0$ for all $i \neq j$.
- The covariance matrix C_{XX} is **diagonal**.
- The joint PDF factorizes into the product of individual Gaussian PDFs:

$$f_{\underline{x}}(\underline{x}) = \prod_{n=1}^{N} \frac{1}{2\pi\sigma_n^2} \exp\left(-\frac{(x_n - \mu_n)^2}{2\sigma_n^2}\right)$$

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4. Worked Examples & Solutions

Numerical Transformation Example

Given $\underline{Y} = \underline{AX}$ with:

$$\underline{\mu}_X = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \underline{A} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}, R_{XX} = \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix}$$

a) Mean Vector $\underline{\mu}_Y$:

$$\underline{\mu}_Y = \underline{A}\underline{\mu}_X = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}.$$

b) Correlation Matrix R_{YY}:

$$R_{YY} = \underline{A}R_{XX}\underline{A}^T = \begin{bmatrix} 10 & -1 & -7 \\ 1 & -7 & 7 \\ -11 & 8 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}.$$

c) Covariance Matrix C_{YY}:
Using $C_{YY} = R_{YY} - \underline{\mu}_Y\underline{\mu}_Y^T$.

1. Outer product: $\underline{\mu}_Y\underline{\mu}_Y^T = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} \begin{bmatrix} 1 & 2 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}.$
2. Subtraction: $\begin{bmatrix} 17 & -6 & -9 \\ -6 & 14 & -8 \\ -9 & -8 & 19 \end{bmatrix} - \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}.$

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5. Case Study: Smart City Traffic (Ahmedabad)

Step 1: Define Variables ($N = 5$)

X_1: Traffic flow, X_2: Avg vehicle speed, X_3: Rainfall intensity, X_4: Temperature, X_5: Air pollution index (AQI).

Step 2: Interpret Covariance (Σ)

- $Cov(X_1, X_3) > 0$: More rain $\implies$ more congestion.
- $Cov(X_2, X_3) < 0$: More rain $\implies$ lower speed.
- $Cov(X_1, X_5) > 0$: More traffic $\implies$ higher pollution.

Step 3: Transformation

Congestion Index $Y = 0.6X_1 - 0.4X_2 + 0.8X_3$.

- If input $\underline{X}$ is Gaussian, Y is also Gaussian.
- Mean: $E[Y] = a^T\mu$.
- Variance: $Var(Y) = a^T\Sigma a$.

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6. Exam Revision Checklist (Homework)

- **Finding Mean:** Given C_{XX} and R_{XX}, solve for $\underline{\mu}_X$ using $\underline{\mu}_X\underline{\mu}_X^T = R_{XX} - C_{XX}$.
- **Variance Proof:** Prove $Var[X+Y] = Var[X] + Var[Y] + 2Cov(X,Y)$.
- **Gaussian PDF:** Be able to calculate the determinant and inverse of a 2×2 covariance matrix to build the joint PDF.

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1. Statistical Representation of Random Vectors

For multiple random variables $X_1, X_2, \ldots, X_N$, we represent them collectively as a **Random Vector** $\underline{X}$.

- **Definition:** $\underline{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$.
- **Purpose:** Provides a compact representation for managing multiple random variables simultaneously.

Mean Vector ($\underline{\mu}_X$)

- **Definition:** The mean vector is the expectation of the random vector: $\underline{\mu}_X = E[\underline{X}]$.
- **Logic:** It represents the average value of the entire random vector.
- **Structure:** Each element corresponds to the mean of an individual random variable within the vector:

$$\underline{\mu}_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix} \quad [\text{cite: } 46]$$

Correlation Matrix (R_{XX})

- **Definition:** $R_{XX} = E[\underline{X}\underline{X}^T]$.
- **Elements:** $R_{XX}(i,j) = E[X_i X_j]$ for $i, j \leq N$.
- **Property:** R_{XX} is a symmetric matrix of size $N \times N$.

Covariance Matrix (C_{XX})

- **Definition:** The "Relationship Map" of random vectors.
- **Formula:** $C_{XX} = E[(\underline{X} - \underline{\mu}_X)(\underline{X} - \underline{\mu}_X)^T]$.
- **Practical Structure:** The matrix contains variances on the diagonal and covariances on the off-diagonals:

$$C_{XX} = \begin{bmatrix} Var(X_1) & Cov(X_1, X_2) & \ldots \\ Cov(X_2, X_1) & Var(X_2) & \ldots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

- **Symmetry:** Because $Cov(X_i, X_j) = Cov(X_j, X_i)$, the matrix is a "mirror" of itself.

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2. Linear Transformations of Random Vectors

Consider the linear model transformation $\underline{Y} = \underline{A}\underline{X} + \underline{b}$, where:

- $\underline{A}$: Transformation matrix (scales and rotates data).

- $\underline{b}$: Constant vector (shifts/translates the center).
- $\underline{X}$: Input random vector.

Mapping Properties Summary

Property	Transformed Result
Mean	$\mu_V = A\mu_X + b$
Covariance	$C_{VV} = AC_{XX}A^T$
Correlation	$R_{VV} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$

Derivations

- **Mean (μ_Y):** Substitute $\underline{Y} = \underline{AX} + \underline{b}$ into $E[\underline{Y}]$. Apply linearity to get $E[\underline{AX}] + E[\underline{b}]$, resulting in $\underline{A}\mu_X + \underline{b}$.
- **Covariance (C_{YY}):** The deviation from the mean is independent of the shift $\underline{b}$: $(\underline{AX} + \underline{b}) - (\underline{A}\mu_X + \underline{b}) = \underline{A}(\underline{X} - \mu_X)$.
  - $C_{YY} = E[\underline{A}(\underline{X} - \mu_X)(\underline{X} - \mu_X)^T \underline{A}^T] = \underline{A}C_{XX}\underline{A}^T$.
  - **Note:** The shift $\underline{b}$ does not affect the covariance (spread).

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3. Multivariate Gaussian Distributions

Dimensions

- **1D (Univariate):** $f_X(x) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.
- **2D (Joint):** Defined by means (μ_X, μ_Y), variances (σ_X, σ_Y), and correlation coefficient (ρ_{XY}).
- **N-Dimensional (General Vector PDF):**

$$f_{\underline{X}}(x) = \frac{1}{(2\pi)^N \det(C_{XX})} \exp\left(-\frac{1}{2}(x - \underline{\mu}_X)^T C_{XX}^{-1}(x - \underline{\mu}_X)\right) \quad \text{[cite: 324, 341]}$$

Uncorrelated Gaussian Random Variables

- **Property:** For mutually uncorrelated variables, $Cov(X_i, X_j) = 0$ for all $i \neq j$.
- **Matrix Structure:** The covariance matrix C_{XX} and its inverse C_{XX}^{-1} are diagonal.
- **Result:** The joint Gaussian PDF factorizes into the product of individual Gaussian PDFs:

$$f_{\underline{x}}(\underline{x}) = \prod_{n=1}^{N} \frac{1}{2\pi\sigma_n^2} \exp\left(-\frac{(x_n - \mu_n)^2}{2\sigma_n^2}\right)$$

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4. Worked Example: Numerical Linear Transformation

Given $\underline{Y} = \underline{AX}$ with:

$$\underline{\mu}_x = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \underline{A} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ -1 & 1 & 0 \end{bmatrix}, \underline{R}_{XX} = \begin{bmatrix} 13 & 2 & 3 \\ 2 & 10 & 3 \\ 3 & 3 & 10 \end{bmatrix}$$

- **(a) Mean vector of Y:** $\underline{\mu}_Y = \underline{A}\underline{\mu}_X = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$.
- **(b) Correlation matrix of Y:** $\underline{R}_{YY} = \underline{A}\underline{R}_{XX}\underline{A}^T = \begin{bmatrix} 17 & -6 & -11 \\ -6 & 14 & -8 \\ -11 & -8 & 19 \end{bmatrix}$.
- **(c) Covariance matrix of Y:** $\underline{C}_{YY} = \underline{R}_{YY} - \underline{\mu}_Y\underline{\mu}_Y^T$.
  - $\underline{\mu}_Y\underline{\mu}_Y^T = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -3 & -6 & 9 \end{bmatrix}$.
  - $\underline{C}_{YY} = \begin{bmatrix} 16 & -8 & -8 \\ -8 & 10 & -2 \\ -8 & -2 & 10 \end{bmatrix}$.

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5. Case Study: Smart City Traffic (Ahmedabad)

System Variables ($N = 5$): X_1: Traffic flow, X_2: Avg speed, X_3: Rainfall intensity, X_4: Temperature, X_5: AQI .

Covariance Interpretation ("Dependency Map"):

- $Cov(X_1, X_3) > 0$: More rain $\implies$ more congestion.
- $Cov(X_2, X_3) < 0$: More rain $\implies$ lower speed. * $Cov(X_1, X_5) > 0$: More traffic $\implies$ higher pollution.
  Congestion Index (Y): Defined as $Y = 0.6X_1 - 0.4X_2 + 0.8X_3$.
- This is a linear transformation ($Y = a^T X$).
- If X is Gaussian, Y is also Gaussian.

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6. Homework Assignments

1. **Finding Mean Vector:** Solve $\underline{\mu}_X\underline{\mu}_X^T = R_{XX} - C_{XX}$ and identify elements from the diagonal.
2. **Variance Proof:** Prove $Var[X + Y] = Var[X] + Var[Y] + 2Cov(X, Y)$.
3. **PDF Derivation:** Compute determinant and inverse of a 2×2 matrix to prove the 2D Gaussian PDF form.

Would you like me to provide the step-by-step matrix inversion derivation for the 2D Gaussian PDF proof found in slides 39-47?